

EE-482: Network Security and Cryptography

LECTURE NO 2: *Model for Network Security*

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Organization of Lecture

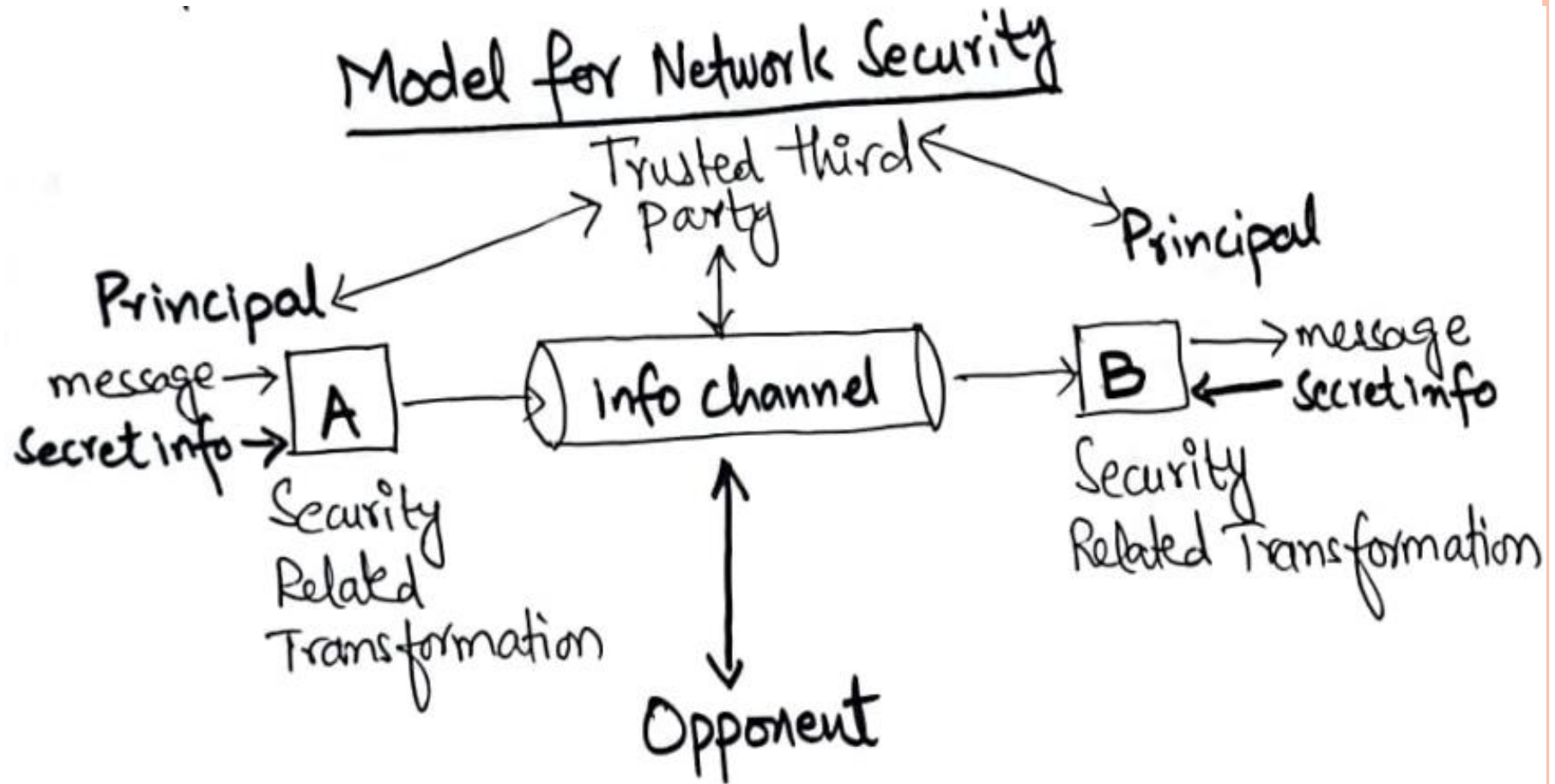
1. Model for Network Security.
2. Network Access Security Model.

1. Model For Network Security (1/5)

Model For Network Security

- ↳ Message is being transferred from 'A' to 'B'.
- ↳ 'A' & 'B' are known as Principals.
- ↳ Communication protocol TCP/IP is followed by both principals.
- ↳ Assume that an opponent (threat) is also present in the system.

1. Model For Network Security (2/5)



1. Model For Network Security (3/5)

↳ All techniques providing security have 3 aspects.

(a) A security related transformation of information:

* Scrambles the message so it is unreadable by opponent

Transformation

PROOF
R00F P

V.S

Substitution

PROOF
QSP PG

Plaintext
Cipher text

→ Encryption of message (various algorithm DES, Ceaser, AES etc)

1. Model For Network Security (4/5)

(b) A code to verify identity of sender:

* To counter authentication attack. Example: Digital Signature

(c) Some secret info. shared by two principals:

* Shouldn't be known to the opponent.

* Secret info is also called **KEY**.

↳ A Trusted Third Party required for secure transmission.

① Responsible for distributing key to the principals.

② Arbitrating disputes b/w the two principals concerning authenticity of a message transmission.

1. Model For Network Security (5/5)

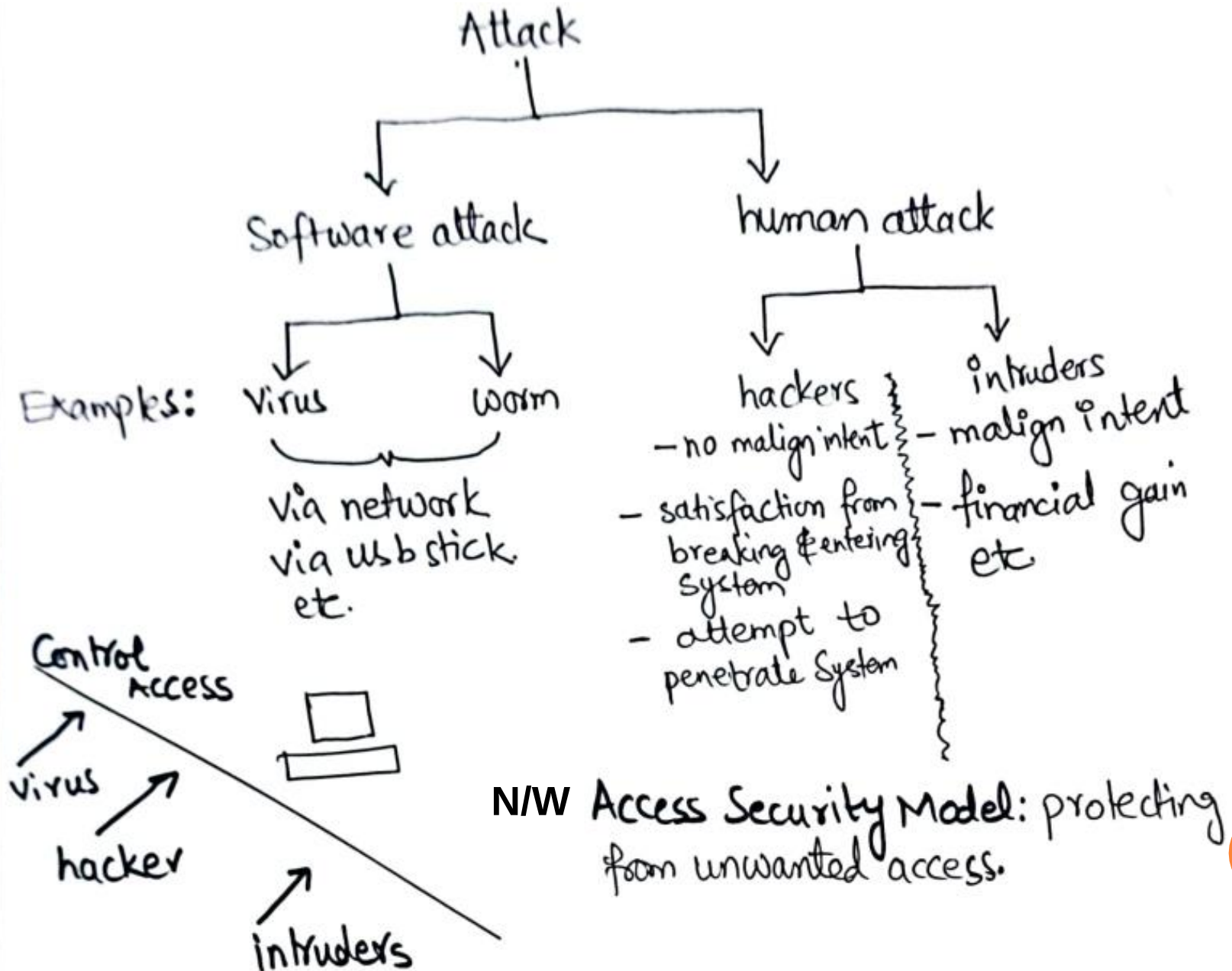
Recall!

Repudiation $\Rightarrow$ $\begin{cases} \text{'A' may deny transmitting message.} \\ \text{'B' may deny receipt of message} \end{cases}$

* 4 Basic Tasks in Designing a Security Service:

- 1) Design an algorithm for performing security related transformation.
- 2) Generate the secret info to be used with the algo.
- 3) Develop method for distribution & sharing of secret info.
- 4) Specify a protocol to be used by the 2 principals that makes use of security algo & secret information.

2. Network Access Security Model (1/3)



2. Network Access Security Model (2/3)

* To cope with unwanted access, 2 categories of security mechanisms

(a) Gate Keeper Function

(b) Internal Security control.

Access Security Model

Opponent

- Human
- Software

Access channel



Gate keeper
function



2. Network Access Security Model (3/3)

(a) Gate Keeper function:

- ↳ includes password-based login procedures.
- ↳ deny access to all but authorized users.
- ↳ screening logic to detect & reject worms, viruses & other software attacks.

(b) Internal Security Controls:

- ↳ Second line of defense
- ↳ Monitoring software to monitor activity & analyze stored info. to detect presence of unwanted intruders.
- ↳ Antivirus softwares to remove malicious programs.

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*